

TerraMaster NAS — Infrastructure Audit & Improvement Plan

Host: terramaster (192.168.0.172) · Ubuntu 24.04.4 LTS

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1. Executive summary

terramaster is a single Intel N95 mini-NAS (4 cores, 7.5 GiB RAM) running Ubuntu 24.04 that has grown organically into a 41-container, 10-Docker-Compose-project home server. It does four unrelated jobs at once:

- 1. Media automation & streaming — Plex + the “arr” stack (Radarr, Sonarr, Prowlarr, qBittorrent) pulling from TorrentLeech and MyAnonaMouse.
- 2. Photo library — Immich (31,975 assets / ~268 GB originals, single user).
- 3. Home-energy automation — the “dash.home2” Sigenergy battery/solar controller (project home-dash2) plus a Life360 family-location feed.
- 4. A cluster of personal web apps & experiments — an algorithmic-trading data platform, a study site, a psychology-revision app, a public marketing site behind a Cloudflare tunnel, a research agent (LiteLLM + SearXNG + GPT-Researcher), a Prometheus/Grafana monitoring stack, and several abandoned prototypes.

It works, but it is fragile and over-exposed. The headline problems:

Area	Problem	Severity
Security	Grafana admin password is admin, exposed on 0.0.0.0:3001	High
Security	~25 service ports bound to 0.0.0.0 with no host firewall (ufw not installed)	High
Security	Plaintext personal passwords & API keys in ~/.env, ~/sigen-dashboard/.env, prowlarr.db, theownitguy-website/.env	High
Security	SSH on 0.0.0.0:22 — PasswordAuthentication yes (verified via sshd -T), PermitRootLogin without-password, fail2ban inactive, and the sudo/login password for tm is literally tm	High
Security	NFS exports /mnt/storage read-write, no auth, to the whole 192.168.0.0/24	Medium

Area	Problem	Severity
Robustness	No UPS ; NVMe SMART shows 35 unclean shutdowns across the two SSDs — ext4 + Postgres + SQLite are being power-cut roughly monthly	Medium–High
Robustness	No SMART self-test schedule ; <code>smartd</code> only mails <code>root</code> locally (i.e. nowhere) — on a no-parity pool this is the blind spot	Medium
Robustness	~ 22 GB real Docker footprint, ~ 38 GB nominal-reclaimable images + 15 GB build cache	Medium
Robustness	<code>home.dash2</code> git repo is in detached HEAD ; most infra config is not in git at all	Medium
Robustness	Pending reboot for kernel 6.8.0-139 (running -138). (<i>Auto-updates via unattended-upgrades are working — earlier draft said otherwise.</i>)	Low–Med
Filesystem	Immich DB (4.7 GB) + model cache live on the OS SSD ; the 932 GB “photos” NVMe (<code>/mnt/NVMe2</code>) is 99 % empty	Medium
Filesystem	<code>/mnt/storage/torrents</code> = 7.6 TB (16 % of the pool) — seeding backlog / copies not hardlinks	Medium
Legacy	6 dead containers, 3 stale compose projects (<code>v2</code> , <code>life360-nas-poller</code> , staging), a failed <code>glancesweb.service</code> , 9 <code>.bundle</code> files, a 4.4 GB stray SQL dump	Low–Med

Nothing here is on fire. The recommendations in §14 are ordered so the security items can be done in an evening and the cleanup over a weekend.

2. Hardware & operating system

Property	Value	Evidence
Model	TerraMaster (F4-style), <code>bond0</code> networking	<code>mount</code> , <code>ss</code> show <code>Interface=bond0</code>
CPU	Intel N95 @ up to 3.4 GHz, 4 cores / 4 threads	<code>lscpu</code>
RAM	7.5 GiB total; at audit 3.7 GiB used, 4.2 GiB cache, 279 MiB free; swap 2.5 GiB / 4 GiB used	<code>free -h</code>
GPU	Intel iGPU (<code>/dev/dri</code>) — used by Plex & Immich for QuickSync transcoding	<code>docker-compose.yml</code> <code>devices</code> :
OS	Ubuntu 24.04.4 LTS (noble), kernel 6.8.0-138-generic	<code>/etc/os-release</code> , <code>uname -a</code>
Uptime	3 days 5 h at audit; load average ~1.0 (i.e. one core pinned continuously)	<code>uptime</code>
Boot	UEFI, <code>curtin</code> /cloud-init install	<code>/etc/fstab</code> comments

Property	Value	Evidence
Pending	<code>/var/run/reboot-required</code> <code>present</code> ; <code>reboot-required.pkgs =</code> <code>linux-image-6.8.0-139-generic</code> (running -138)	<code>cat</code> <code>/var/run/reboot-required.pkgs</code>
Auto-updates	<code>unattended-upgrades is enabled</code> <code>and running</code> — <code>/etc/apt/apt.conf.d/20auto-upgrades</code> sets both <code>Update-Package-Lists</code> <code>"1"</code> and <code>Unattended-Upgrade "1"</code> ; <code>apt-daily-upgrade.timer</code> fired 2026-09-05 06:04 and patched <code>openssh</code> , <code>gnupg</code> , <code>linux-* 138->139</code> , <code>libcrypt</code> , etc.	<code>/var/log/apt/history.log</code>

Observation: swap is 62 % used with 4 GiB cache in RAM — the box is under steady memory pressure. `immich` (386 MB RSS), two `Immich Postgres` instances, `Radarr/Sonarr/Prowlarr` (.NET, ~500 MB combined), `Grafana`, `Prometheus`, the 7-container trading platform, and `home-dash2` all resident. Load average pinned at ~1.0 is `trading-collector` (10 % CPU continuously) plus `Immich ML`. **Reassuring:** `journalctl -k -b` over 3+ days shows **no OOM kills, no I/O errors, no ext4 errors, no ATA resets** — swap is absorbing the pressure without anything being killed. Only boot-time noise: cosmetic ACPI-BIOS symbol errors (N95 firmware quirk), a `blkmapd` NFS pipe warning, and the 2nd NIC (`enp2s0`) link-down (unplugged).

2.1 Disk & power health (SMART, verified with `sudo`)

Every drive passes **SMART** with *zero* reallocated sectors, *zero* pending sectors, *zero* offline-uncorrectable, *zero* **UDMA-CRC** errors. The pool is genuinely healthy today. The caveats are age, drive type, and the lack of monitoring/parity/UPS.

Dev	Mount	Model	Type	Power-on hours	Realloc / Pending / CRC	Temp	Note
sda	disk1	WD60EFAX-68JH4N0	SMR (WD Red SMR), 5400 rpm	4,991	0 / 0 / 0	24 °C	SMR is poor for the torrent-move workload Oldest by far — statistically first to fail; holds ~4.2 TB, no parity healthy
sdb	disk2	WD60EFRX-68MYMN1	CMR (WD Red), 5700 rpm	45,608 (~5.2 yr)	0 / 0 / 0	31 °C	
sdc	disk3	WD60EFRX-68MYMN1	CMR, 5700 rpm	14,334	0 / 0 / 0	33 °C	
sdd	disk4	WUH721818ALECMR	CMR (Ultrastar DC HC550), 7200 rpm	31,514	0 / 0 / 0	36 °C	healthy
sde	disk5	WUH721818ALECMR	CMR, 7200 rpm	31,379	0 / 0 / 0	37 °C (max-ever 49 °C)	slightly warm
sdf	disk6	WUH721818ALECMR	CMR, 7200 rpm	21,835	0 / 0 / 0	37 °C	healthy

Dev	Mount	Model	Type	Power-on hours	Realloc / Pending / CRC	Temp	Note
nvme0	/mnt/NVMe2	WD SN550 1TB	TLC	4,611	wear 6 % , 35 TB written, 21 unsafe shutdowns , 0 media errors	—	the under-used “photos” NVMe
nvme1	/ (OS)	WD SN550 1TB	TLC	4,911	wear 1 % , 23 TB written, 14 unsafe shutdowns , 0 media errors	—	OS + Immich DB + Plex metadata

Three robustness gaps fall out of this:

1. **No UPS.** `lsusb` shows no UPS; no `nut/apcupsd` installed. The NVMe counters record **35 combined unsafe shutdowns** — the machine is losing power uncleanly on the order of once a month. `ext4`, `mergerfs`, `Postgres` (Immich + trading), and a dozen `SQLite` databases are all exposed to that.
2. **No SMART self-test schedule.** `/etc/smartd.conf` is only `DEVICESCAN -d removable -n standby -m root -M exec ...` — no `-s (S/.../.../02|L/.../.../6/03)` short/long test schedule, and alerts go to local root mail (unread). On a **no-parity** pool, undetected drive degradation is the worst case.
3. **The 5.2-year-old disk2 (sdb).** Still clean, but this is the one to pre-emptively replace or at least watch closely — and the SMR `disk1 (sda)` is the one whose write performance will degrade under load.

3. Storage architecture

3.1 Physical disks

```
lsblk (abridged):
sda  5.5T WD60EFAX  ext4  /mnt/disk1
sdb  5.5T WD60EFRX  ext4  /mnt/disk2
sdc  5.5T WD60EFRX  ext4  /mnt/disk3
sdd  16.4T WUH721818 ext4  /mnt/disk4
sde  16.4T WUH721818 ext4  /mnt/disk5
sdf  16.4T WUH721818 ext4  /mnt/disk6
nvme0n1  931.5G WD SN550  xfs   /mnt/NVMe2      (fstab: "The Second NVMe2 for Photos")
nvme1n1  931.5G WD SN550  ext4  /               + /boot/efi      (OS)
```

3.2 mergerfs pool

`/etc/fstab`:

```
/mnt/disk* /mnt/storage mergerfs defaults,allow_other,cache.files=off,use_ino,\
category.create=mfs,minfreespace=100G,fsname=mergerfsPool,nonempty 0 0
```

- **6 data disks (66 TB raw)** pooled as `/mnt/storage` via `mergerfs` (FUSE union).
- `category.create=mfs` = new files go to the disk with **most free space**.
- **There is no parity / RAID.** `mergerfs` is JBOD-with-a-union-view: lose one disk, lose exactly the files on that disk. `mergerfs.balance` and `mergerfs-tools` (cloned from `trapexit/mergerfs-tools`) are present for rebalancing.
- Pool usage: **47 TB used / 66 TB (75 %)**, 16 TB free.

`df -h` at audit:

Mount	Size	Used	Use %
/ (OS SSD)	915 G	99 G	12 %
/mnt/storage (pool)	66 T	47 T	75 %
/mnt/disk1	5.5 T	4.0 T	77 %
/mnt/disk2	5.5 T	4.2 T	78 %
/mnt/disk3	5.5 T	4.0 T	77 %
/mnt/disk4	17 T	12 T	78 %
/mnt/disk5	17 T	12 T	78 %
/mnt/disk6	17 T	11 T	67 %
/mnt/NVMe2	932 G	58 G (2.8 G real; xfs overhead)	7 %

3.3 What lives where (logical map)

Path	Contents	Size	Notes
/mnt/storage/plex-movies	Radarr library root, Plex “Movies”	26 TB	1,137 folders / 1,127 Plex items
/mnt/storage/plex-tv	Sonarr library root, Plex “TV Shows”	12 TB	632 show folders / 5,763 episodes
/mnt/storage/plex-mobile	Plex “Mobile-Movies” (re-encoded small copies)	922 GB	930 folders
/mnt/storage/torrents	qBittorrent completed-download area for radarr/sonarr	7.6 TB	688 entries — see §5.4
/mnt/storage/jez-photos	Immich upload root (/usr/src/app/upload)	~300–350 GB	268 GB originals per Immich DB + thumbs/encoded
/mnt/storage/F1	qBittorrent “F1” category + Plex “F1” library	(in pool)	also mis-mapped to /mnt/NVMe2 in Plex
/mnt/storage/EPL, /mnt/storage/data	sports / misc download landing	small	
/mnt/storage/homes	Samba [homes] per-user dirs	small	root-owned
/mnt/storage/research-agent	app data on the pool		
/mnt/storage/plex-mobile			
/jez-cache/immich/postgres	Immich Postgres data dir	4.7 GB on disk (2.2 GB logical)	on the OS SSD /, not NVMe2
/jez-cache/immich/model-cache	Immich ML models	786 MB	on the OS SSD
/var/lib/docker	images / overlay / volumes	22 GB	~half reclaimable (§13)
/mnt/NVMe2/trading-platform	trading Postgres + parquet	2.8 GB	one of only two real users of the “photos” NVMe source
/mnt/NVMe2/family-home-backup	highly snapshot of /mnt/storage/homes, via family-home-backup.timer (02:15) -> /usr/local/sbin/backup-family-homes	~0	/mnt/storage/homes is itself only 48 KB — feature set up 2026-08-31, barely used
/home/tm/config/Library	Plex metadata / thumbnails / DB	31 GB	on the OS SSD (fine, but large)
/home/tm/config/{radarr,sonarr,plex-tv,plex-movies}	meta-configs + debug logs	2.3 G / 1.2 G / 1.2 G / 422 M	log level is debug everywhere — inflated
/mnt/immich_db_backup.sql	stray 4.4 GB manual dump, 2026-07-04	4.4 GB	legacy — delete

Two filesystem problems jump out:

1. **The 932 GB NVMe (/mnt/NVMe2) that fstab labels “for Photos” is 99 % empty.** Immich’s database and ML cache sit on the OS SSD instead, and the photo originals sit on spinning mergerfs disks. Immich thumbnail generation and timeline scrubbing are exactly the random-IO workload an NVMe is for.
2. **/mnt is a junk drawer:** immich_db_backup.sql (4.4 GB), temp/, scratch/, nvme1n1p2-scratch/, usb_data/, remote_usb/, das1/, das2/ (empty DAS mountpoints), and .DS_Store files committed all over /mnt/storage.

4. Docker landscape

docker info: 47 containers (41 running), 49 images, overlay2, root /var/lib/docker. docker system df:

```
Images           49    ACTIVE 42    SIZE 39.11GB    RECLAIMABLE 38.32GB (97%)
Containers       47    ACTIVE 41    SIZE 268.3MB
Local Volumes    8     ACTIVE 6     SIZE 326.1MB
Build Cache      832                               SIZE 17.26GB    RECLAIMABLE 14.92GB
```

4.1 Compose projects (docker compose ls -a)

Project	Config file	State	Purpose
tm	~/docker-compose.yml + ~/docker-compose-monitoring.yml	19 running, 1 exited	Media stack + Immich + Plex + monitoring
trading-platform	~/trading-platform/deploy/docker-compose.terraform.yml	ASX1	crypto data platform
home-dash2	~/sigen-dashboard/home.dash2/docker-compose.yml	2 running	dash.home2 energy dashboard
research-agent	~/research-agent/docker-compose.yml	3 running	LiteLLM + SearXNG + GPT-Researcher
territory	~/territory/docker-compose.yml	2 running	news-signal watcher web app
theownitguy-website	~/theownitguy-website/docker-compose.yml	2 running	public marketing site + Cloudflare tunnel
timeline-nas-webapp	~/timeline-nas-webapp/docker-compose.yml	1 running	Google-Takeout family timeline
vce-psychology	~/vce-psychology/docker-compose.yml	1 running	VCE Psychology revision app (ap.study)
v2	~/sigen-dashboard/v2/docker-compose.yml	1 exited (3)	superseded predecessor of home-dash2
life360-nas-poller	~/life360-nas-poller/docker-compose.yml	1 exited (2)	standalone Life360 app, replaced by home-dash2’s built-in poller

4.2 Docker networks

18 networks. Live/normal: tm_default, tm_immich-network, tm_monitoring, home-dash2_default, research-network, territory_default, trading-platform_trading_internal, dashboard_shared (shared trading<->dash), theownitguy-website, timeline-nas-webapp_default, vce-psychology_default.

Orphaned: homedash2_default, home-dash2_default (duplicate naming — home.dash2 vs homedash2), life360-nas-poller_default, v2_default.

4.3 Container privilege posture

- No container is --privileged. [OK]
- No container mounts /var/run/docker.sock. [OK]
- 6 containers use network_mode: host: plex2, qbt-tl, sonarr, radarr, prowlarr, autobrr. Host networking is normal for Plex (DLNA/GDM discovery) but means Radarr:7878, Sonarr:8989, Prowlarr:9696 and the qBittorrent WebUI:8080 are directly on the LAN with no Docker port mapping to restrict them.

- The **trading-platform** stack is the model citizen: `read_only: true`, `cap_drop: [ALL]`, `no-new-privileges`, `tmpfs` for `/tmp`, Docker *secrets* for the DB password, `mem_limit/cpus` on every service, all ports bound to `127.0.0.1`. Nothing else on the box is this disciplined.

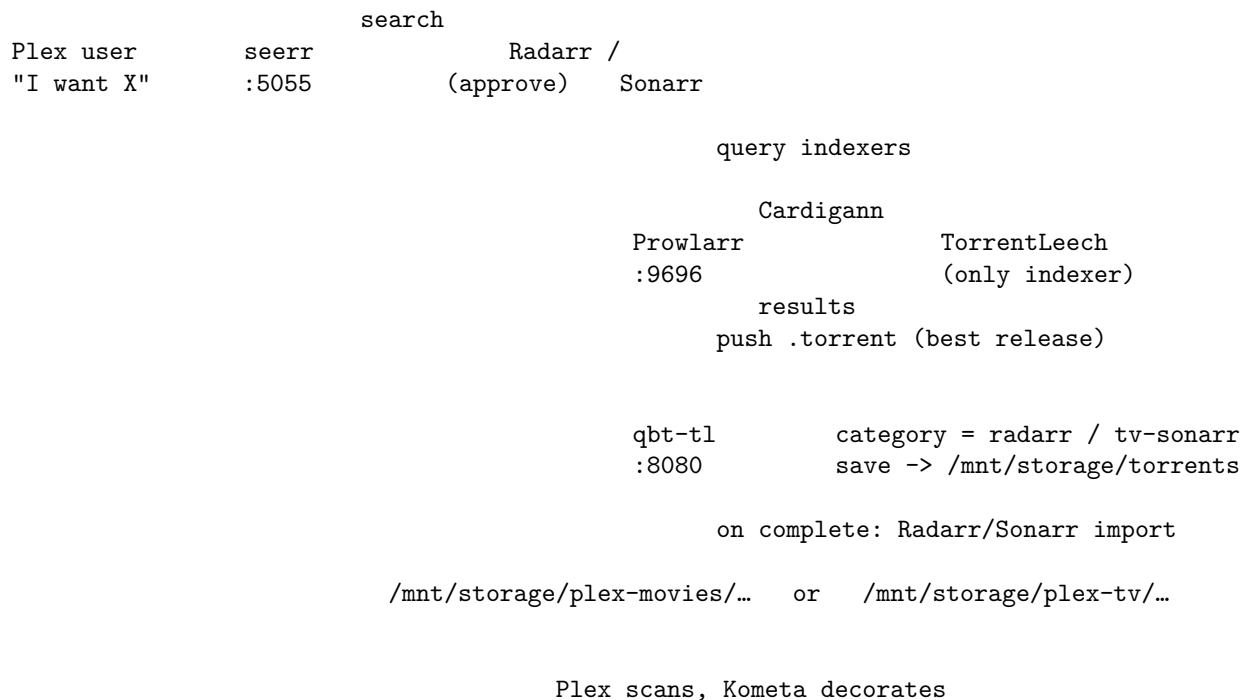
5. Media automation stack (the “*arr*” stack)

5.1 Components

All defined in `~/docker-compose.yml` (project `tm`), images from `lscr.io/linuxserver/*`, `PUID=1000 PGID=1000`, `TZ=Australia/Melbourne`, `/mnt` bind-mounted whole into each:

Container	Image	Port	Config dir	Role
<code>prowlarr</code>	<code>linuxserver/prowlarr</code>	9696 (host)	<code>~/config/prowlarr</code>	Indexer manager — the single source of “where to search”
<code>radarr</code>	<code>linuxserver/radarr</code>	7878 (host)	<code>~/config/radarr</code>	Movie automation
<code>sonarr</code>	<code>linuxserver/sonarr</code>	8989 (host)	<code>~/config/sonarr</code>	TV automation
<code>qbt-tl</code>	<code>linuxserver/qbittorrent</code>	8080 (host)	<code>~/config/qbt-tl</code>	The one torrent client
<code>autobrr</code>	<code>ghcr.io/autobrr/autobrr</code>	7474 (host)	<code>~/config/autobrr</code>	IRC announce -> instant grabs — installed but empty
<code>seerr</code>	<code>ghcr.io/seerr-team/seerr</code>	5055	<code>~/config/overseerr</code>	Request front-end (Overseerr fork)
<code>kometa</code>	<code>kometateam/kometa</code>	—	<code>~/config/kometa</code>	Plex collection/artwork manager, runs 03:00 daily

5.2 How a download actually happens



Evidence

Prowlarr (prowlarr.db):

- **Indexers (1):** TorrentLeech — Implementation: Cardigann, definitionFile: torrentleech, credentials embedded (see §11).
- **Applications (2):** Radarr (<http://localhost:7878>) and Sonarr (<http://localhost:8989>), both SyncLevel: 2 (“Full Sync” — Prowlarr owns the indexer list in Radarr/Sonarr).
- **Download client (1):** qBittorrent localhost:8080, category prowlarr.

Radarr (radarr.db):

```
RootFolders:      /mnt/storage/plex-movies/
DownloadClients:  qBittorrent localhost:8080 user=qbittorrent pass=qbittorrent
                  movieCategory = "radarr"
Config:  autounmonitorpreviouslydownloadedmovies=True
         deleteemptyfolders=True  minimumfreespacewhenimporting=1000 (MB)
```

Sonarr (sonarr.db):

```
RootFolders:      /mnt/storage/plex-tv/
DownloadClients:  qBittorrent localhost:8080 user=qbittorrent pass=qbittorrent
                  tvCategory = "tv-sonarr"
Config:  enablecompleteddownloadhandling=True  deleteemptyfolders=True
```

qBittorrent categories (~/.config/qbt-t1/qBittorrent/categories.json):

Category	Save path
radarr	/mnt/storage/torrents
tv-sonarr	/mnt/storage/torrents
prowlarr	<i>(empty — uses default)</i>
new movies	/mnt/storage/data/movies
new tv / tv	/mnt/storage/data/tv
F1	/mnt/storage/F1
Books	/mnt/disk5/data/books
remoteT5	/mnt/remote_usb/T5

qBittorrent session config (qBittorrent.conf):

```
Session\DefaultSavePath=/mnt/storage/data
Session\TempPathEnabled=false
Session\Port=40012          Session\Interface=bond0  InterfaceAddress=192.168.0.172
Session\DHTEnabled=false  PeXEnabled=false  LSDEnabled=false  + private-tracker safe
Preferences\WebUI\Address=*  WebUI\Username=qbittorrent
Preferences\Downloads\SavePath=/mnt/disk2/temp/      + STALE, contradicts DefaultSavePath
```

5.3 Notable configuration facts

- **Authentication on Radarr/Sonarr/Prowlarr = “Forms” but AuthenticationRequired = DisabledForLocalAddress** — i.e. **no login at all from the LAN**. Anyone on 192.168.0.0/24 has full admin.
- API keys are in cleartext in each config.xml (see §11).
- LogLevel = debug on all three — this is why ~/.config/radarr is 2.3 GB.
- **autobrr is running but 100 % unconfigured:** autobrr.db has 0 users, 0 indexers, 0 filters, 0 IRC networks, 0 actions. It is dead weight and an open :7474 port. Either wire it to TorrentLeech’s IRC announce channel (its actual purpose — near-instant grabs) or remove it.
- Only **one** qBittorrent instance despite the compose header comment (“qbt-sports is for downloading F1, AFL, EPL from Sports Cult”). The qbt-sports container referenced in comments does not exist; F1 is handled by categories in qbt-t1.

5.4 The 7.6 TB /mnt/storage/torrents problem

/mnt/storage/torrents holds **7.6 TB across 688 entries** — 16 % of the whole pool. Radarr and Sonarr import from there into `plex-movies` / `plex-tv`. Both source and destination are under the same `mergerfs` mount, so **hardlink imports are possible only when Radarr/Sonarr’s “Use Hardlinks instead of Copy” is on and mergerfs keeps both paths on the same underlying disk**. Given the size, a large fraction of this is either:

- duplicated (copied, not hardlinked) into the library — pure waste, or
- seeding torrents deliberately retained for TorrentLeech ratio, or
- orphaned downloads Radarr/Sonarr failed to import and never cleaned.

This needs a manual reconciliation pass (see Recommendation 15).

6. TorrentLeech & MyAnonaMouse

6.1 TorrentLeech (movies / TV)

- **Reached only through Prowlarr’s Cardigann torrentleech definition.** Account `alchemist168`, password + alt-2FA token stored in `prowlarr.db Indexers.Settings.extraFieldData` (see §11).
- Prowlarr full-syncs the TL indexer into Radarr and Sonarr. Neither *arr* talks to TL directly — they ask Prowlarr, Prowlarr scrapes TL, returns results, the *arr* picks a release by its quality profile and hands the `.torrent` to qBittorrent.
- **qBittorrent is configured private-tracker-safe:** DHT / PeX / LSD all disabled, bound to `bond0` / `192.168.0.172`, port `40012`, UPnP off, `PortForwardingEnabled=false`.
- **No VPN and no bind-fail “kill switch”.** `Proxy\Profiles\*=true` is set but **no proxy host is configured**, so that setting is inert — torrent traffic egresses on the home WAN IP. For private trackers this is the normal and expected setup; noted only so it is a conscious choice.
- `autobrr` was clearly intended to add TL’s IRC announce channel for instant grabbing but was never set up.

6.2 MyAnonaMouse (books) — the `mam-check.sh` cron

Separate from the *arr* stack:

- `crontab -l` (user `tm`): `0 3,9,15,21 * * * /bin/bash /home/tm/scripts/mam-check.sh`
 - `~/scripts/mam-check.sh` logs into MyAnonaMouse with a hard-coded MAMID session token, keeps a session cookie in `~/mam_script/MAM.cookies`, and **auto-spends bonus points**: buys VIP, keeps upload credit topped up (100 GB / 50 GB tranches) while staying above a 10,000-point buffer.
 - Books land via the qBittorrent Books category -> `/mnt/disk5/data/books`.
 - The MAMID token in the script is a **long-lived credential in plaintext** in a world-readable-by-`tm` script (see §11).
-

7. Plex

7.1 Setup

`~/docker-compose.yml` service `plex` -> container `plex2`, `lscr.io/linuxserver/plex:latest`, `network_mode: host`, `devices: /dev/dri:/dev/dri` (QuickSync).

Config path: the compose file mounts `./config:/config`, and `./` is `/home/tm`, so Plex’s `/config` is the entire `~/config` directory. Plex’s own data therefore lives at:

`/home/tm/config/Library/Application Support/Plex Media Server/` (31 GB)

This is messy — Plex shares a config root with Radarr, Sonarr, Prowlarr, Kometa, etc. It works because each app uses its own subfolder, but a `chown -R` or a backup script pointed at `~/config` now sweeps up 31 GB of Plex thumbnails.

`Preferences.xml` highlights:

```
FriendlyName="TM"
PlexOnlineUsername="Jeremy_Poon"    PlexOnlineMail="jeremy@poons.com"
```

```
PlexOnlineToken="***" (present in cleartext in the file)
TranscoderQuality="2" TranscoderThrottleBuffer="6000"
HardwareDevicePath="8086:46d2...00000:00:02.0" ← HW transcode enabled
ScannerLowPriority="1" GenerateBIFBehavior="scheduled"
DlnaEnabled="0"
```

Transcode temp dir: compose mounts `/mnt/temp:/temp` (host `/mnt/temp`, 24 KB — effectively unused / transcodes to default location).

Two extra Plex-related scripts in `~/scripts/`: `plex_limit.sh` / `plex_resume.sh` (bandwidth throttling helpers, 188 / 77 bytes).

7.2 Libraries (from `com.plexapp.plugins.library.db`)

#	Library	Type	Path(s)	Items
1	Movies	movie	<code>/mnt/storage/plex-movies</code>	1,127
2	TV Shows	show	<code>/mnt/storage/plex-tv</code>	5,763 (episodes+seasons)
3	Mobile-Movies	movie	<code>/mnt/storage/plex-mobile</code>	924
6	F1	movie	<code>/mnt/storage/F1</code> and <code>/mnt/NVMe2</code>	20

- **Where movies are stored:** `/mnt/storage/plex-movies` — the mergerfs pool, physically spread across disk1–disk6 (every disk has a `plex-movies/` subtree). ~**26 TB**, 1,137 on-disk folders.
- The **F1 library is mis-configured**: it points at both `/mnt/storage/F1` and the bare `/mnt/NVMe2` root. That second path is almost certainly a leftover mistake (NVMe2 is supposed to be photos) and should be removed from the library.
- The Plex DB file is **206 MB** and was last written seconds before the audit query — healthy and active.

8. Immich (photo library)

8.1 Topology

`~/docker-compose.yml` (project `tm`), 5 containers on `tm_immich-network`, `env_file: .env`:

Container	Image	Purpose	Storage
<code>immich_server</code>	<code>ghcr.io/immich-app/immich-server-web</code>	API server, port 2283	upload -> <code>/mnt/storage/jez-photos</code> ; internal <code>/data</code> on a Docker volume
<code>immich_machine_learning</code>	<code>.../immich-machine-learning-fuse</code>	for CLIP/OCR models	<code>/jez-cache/immich/model-cache</code> (786 MB, OS SSD)
<code>immich_postgres</code>	<code>ghcr.io/immich-app/postgres</code>	DB (postgresql)	<code>/mnt/storage/immich/postgres</code> (OS SSD), <code>DB_STORAGE_TYPE='HDD'</code>
<code>immich_redis</code>	<code>redis:6.2-alpine</code>	queue	ephemeral
<code>immich_power_tools</code>	<code>ghcr.io/varun-raj/immich-power-tools</code>	background jobs, port 8001	talks straight to the DB

8.2 Library facts (from the live database)

```
assets total ..... 31,975 (29,246 IMAGE + 2,729 VIDEO)
external libraries ... 0 (all assets are uploaded/internal)
users ..... 1 jeremy@poons.com "Jeremy Poon"
originals size ..... 268 GB (sum of asset_exif.fileSizeInByte)
DB size ..... 2,233 MB
capture date range ... 1900-01-01 -> 2026-09-02 ← some assets have broken EXIF dates
```

/mnt/storage/jez-photos/ subfolders: library/, upload/, thumbs/, encoded-video/, profile/, takeout/ (Google Photos import staging), backups/.

8.3 Backups — this part is done right

/mnt/storage/jez-photos/backups/ contains Immich’s **native nightly DB dump**:

```
immich-db-backup-20260822T020000-v3.0.1-pg14.19.sql.gz    ~253 MB
```

... one per night ...

```
immich-db-backup-20260906T020000-v3.0.1-pg14.19.sql.gz
```

14 retained, ~250 MB each, generated 02:00, written 12:01. This is Immich’s built-in `BACKUP_DATABASE` feature and it is working. The dumps are the DB only — the 268 GB of originals are protected only by the mergerfs disk they happen to sit on (no parity, no offsite).

The stray `/mnt/immich_db_backup.sql` (4.4 GB, uncompressed, 2026-07-04) is a one-off manual dump from before the nightly job existed. Delete it.

8.4 Notes

- The `:v3` image tag is unusual (Immich normally tags `release / vX.Y.Z`); the backup filenames say `v3.0.1`, so this is a real Immich 3.x line.
- `DB_PASSWORD=immich`, `JWT_SECRET=jez_power_tools_2026_security` — trivially guessable, and `Postgres:5432` is only inside `tm_immich-network` (not published), so exposure is limited, but `Power-Tools:8001` is on `0.0.0.0`.

9. “dash.home2” — the home-energy dashboard

9.1 What “dash.home2” is

There is no `.home2` hostname anywhere. “**dash.home2**” is your shorthand for the **home-dash2 compose project**, whose code lives in `~/sigen-dashboard/home.dash2/` (a git repo, `origin = git@github.com:alc168/home.dash2.git`). The “2” is an internal generation marker — it is the **second full rewrite** of the Sigenenergy dashboard (`_legacy_v1_unused/` -> `v2/` -> `home.dash2/`).

It is reached in a browser at `https://home/`, `https://home.energy/`, or `https://home.dash/` — nginx (`/etc/nginx/sites-enabled/sigen-dashboard`) terminates TLS with the self-signed `home.energy` cert and reverse-proxies to `127.0.0.1:3000`.

9.2 What it does (from its README)

Every 5 minutes it:

1. Reads the **Sigenenergy SigenStor** inverter over Modbus/TCP (SOC, solar, load, grid flow, charge limits).
2. Pulls wholesale prices from **Amber Electric** for the next 48 half-hours.
3. Predicts whether the battery will be full before the next price peak.
4. If not, and power is cheap now, **charges the battery from the grid**.
5. Renders one web page and sends **WhatsApp alerts** (via a Baileys sidecar) when something looks wrong.

It also reads a **Govee** sauna temp sensor, a **Meross** smart plug (surplus- solar diversion to a Bluetti battery), **UniFi** client counts, and BOM weather; and it exposes a **family-location view** fed by the Life360 poller.

9.3 Runtime (`~/sigen-dashboard/home.dash2/docker-compose.yml`)

Service	Command	Ports	Notes
sigen-web	<code>python -m sigen_dashboard.web</code>	<code>0.0.0.0:3000->3000</code>	the dashboard UI + control API
sigen-scheduler	<code>python -m sigen_dashboard.scheduler</code>	—	the 5-minute decision loop
sigen-baileys	<code>Node/Baileys</code>	<code>0.0.0.0:3002->3001</code>	WhatsApp gateway

Service	Command	Ports	Notes
life360-poller	Dockerfile.life360-poller —		polls Life360's unofficial API, writes <code>life360.db</code>
life360-api	Dockerfile.life360-api	expose 4000 (internal only)	sanitised read API for <code>sigen-web</code>

- Persisted state -> `~/sigen-dashboard/data/` (bind mount, **root-owned**): `history_v2.db` (30 MB, actively growing — the energy time-series), `baileys_auth/` (WhatsApp login), `sauna_config.json`, `topup_config.json`, `known_people.json`, `solar_history*.jsonl`.
- `ps` showed `python -m sigen_dashboard.web` (PID 3345) and `.scheduler` (PID 3162) as **root** — that is the container processes seen from the host PID namespace, **not** a second native copy. There is no systemd unit for this app.
- Config comes from `~/sigen-dashboard/.env` (shared with the retired v2 project) and `~/sigen-dashboard/life360.env`. Both contain **cleartext personal passwords** (see §11).

9.4 Orphaned code around dash.home2

`~/sigen-dashboard/` is the single messiest directory on the box. What is actually in use is **only** `home.dash2/` and the shared `data/`, `.env`, `life360.env`. Everything else is dead:

Path	What it is	Status
<code>_legacy_v1_unused/</code>	the original Flask app.py (63 KB) + a 195 MB history.db + <code>secrets.yaml</code>	dead — keep the DB if it has history you want, else delete
<code>v2/</code>	previous rewrite, compose project v2 still registered, containers sigen-web/sigen-scheduler/sigen-baileys exited 12 days ago	dead — <code>docker compose -p v2 down</code> , delete dir
<code>migration-test/</code>	a copy of the codebase used for a migration dry-run	dead
<code>data-staging/</code> , <code>home.dash2/docker-compose.staging.yml</code>	staging variant	unused
<code>backups/</code> (codex-*, pre-git-cutover-*)	snapshot dirs from before git	superseded by git
Root-level *.md / *.pdf (Architecture, Tutorials, Security Report, ...)	~15 docs, many duplicated in <code>docs/</code> and again as <code>._*</code> AppleDouble files	consolidate
<code>sigen-cron.log</code> , <code>sigen-logger.log</code> in <code>~/</code>	logs from a retired native-cron version of the scheduler	delete
<code>glancesweb.service</code> (systemd, enabled, failed, crash-looped, gave up 2026-09-03)	old Glances web UI on 0.0.0.0 as root	disable + delete unit
<code>data/glances_key*</code>	leftover Glances auth keys	delete

Docker images still on disk for these: `v2-sigen-*`, `home-dash2-staging-*`, `homedash2-sigen-scheduler`, `sigen-map-test-*` — ~3.5 GB.

10. Other application stacks

Stack	Ports	What it is	Storage	Health
trading-platform	127.0.0.1:8092 api, :8093 ui, :9108-9110 metrics	7-service ASX/crypto data & signal platform (collector, api, ui, maintenance, sentiment, macro, equity) + postgres:17	/mnt/NVMe2/trading-platform	healthy; {postgres,parquet} best-hardened stack on the box (read-only, cap-drop, secrets, limits)
research-agent	:4000 litellm, :8088 searxng, :8000 gpt-researcher	Zero-cost research pipeline: LiteLLM gateway -> free Gemini/Groq/Cerebras keys, SearXNG metasearch, GPT-Researcher orchestrator. Proxied at https://research.agent	~/research-agent/*	running; API keys in ~/research-agent/.env
territory	:8801 backend, :8802 frontend	News-signal watcher (Google News / GDELT / press releases), SQLite	~/territory/backend/data	running; git remote is a local bare repo only
vce-psychology	:8137 (-> https://ap.study)	VCE Psychology revision web app, SQLite + flashcard scheduler, optional AI	Docker volume vce-psychology_vce-data; config /opt/vce/config or ~/vce-config	healthy
theownitguy-website	none local — Cloudflare tunnel	Public marketing site (theownitguy), served to the internet by cloudflared on the ownitguy-tunnel network	image build	running; only publicly-reachable service
timeline-nas-webapp	:4002	Family timeline from Google Takeout	~/timeline-nas-webapp/data	running
bitmappery	:5173	Self-hosted image editor (built from github igorski/bitmappery , 3.15 GB image)	Docker volume	running; bitmappery-init-1 exited(0) — normal
Monitoring (~/docker-compose-monitoring.yml)	:9090 prometheus, :9093 alertmanager, :9130 unpoller, node-exporter	Prometheus (365-day retention) + Grafana + Alertmanager + UniFi Poller	~/data/{prometheus,alertmanager,grafana} ~/config/grafana	running; Grafana admin password = admin

11. Exposed secrets inventory

These were found in cleartext during the audit. Treat every one as compromised-if-the-host-is and rotate. Values below are **masked** — the point is *where they live*, so you can fix them.

Secret	Location	Value (masked)	Risk
Grafana admin password	~/docker-compose-monitoring/ -> ~/.env GF_SECURITY_ADMIN_PASSWORD	mdymnl	Critical — full Grafana on 0.0.0.0:3001
Sigen cloud password	~/sigen-dashboard/.env SIGEN_CLOUD_PASSWORD	P15q...n0!	High — appears to be a reused personal password
Meross account password	~/sigen-dashboard/.env MEROSS_PASSWORD	Athe...e5!	High — reused personal password
TerraMaster admin password	~/sigen-dashboard/.env TERRAMASTER_PASSWORD	tm	High — NAS admin is tm/tm
TorrentLeech password + 2FA token	~/config/prowlarr/prowlarr/ (Indexers.Settings)	7atb...8! / af5e...dd	Med — tracker account takeover
MyAnonaMouse session token (MAMID)	~/scripts/mam-check.sh (hard-coded)	DfJz...oD (270 chars)	Med — long-lived, auto-spends points
Life360 bearer token	~/sigen-dashboard/life360/ LIFE360_ACCESS_TOKEN	Bearer NjU4...Vm	Med — family location data
Amber Electric API key	~/sigen-dashboard/.env AMBER_API_KEY	psk_9e24...c2	Low-Med
Govee API key	~/sigen-dashboard/.env GOVEE_API_KEY	ef97...d9a9	Low
UniFi API key + prometheus account pw	~/sigen-dashboard/.env, ~/.env UP_UNIFI_DEFAULT_PASS	Jeov...gK / mr8S...lc	Med — network controller
Immich API key / JWT secret / DB pw	~/.env	Zrr...4g / jez_power_tools_2026_security / immich	Med
Plex claim token	~/.env PLEX_CLAIM	claim-53G4...	Low (claim tokens expire in 4 min)
Plex online token	.../Plex Media Server/Preferences.xml	present	Med — Plex account access
Cloudflare Tunnel token	~/theownitguy-website/.env and visible in docker inspect cloudflared args	veyJ...Jm	Med — anyone who can read it can run your tunnel
Gemini / Groq / Cerebras keys	~/research-agent/.env	AQ.Ab8... / gsk_zg... / csk-tx...	Low-Med
autobrr session secret	~/config/autobrr/config.toml	7B3e...a4	Low (service is unconfigured)
arr API keys	~/config/{radarr,sonarr,prowlarr}/ conf/1g94m1	pr280...01 / cbb71g94m1 fe2e...33	Low (LAN only)

12. Security review

#	Finding	Detail / evidence	Severity
S1	No host firewall (verified)	iptables -S: -P INPUT ACCEPT; the only INPUT rules are Tailscale's ts-input chain. nft list ruleset confirms — no LAN filtering at all. ufw not installed. Every 0.0.0.0 port is reachable from the whole /24.	High

#	Finding	Detail / evidence	Severity
S2	Grafana admin/admin on 0.0.0.0:3001	~/.env; GF_USERS_ALLOW_SIGN_UP=false is the only mitigation	High
S3	Cleartext personal passwords in .env files	§11; several look reused across accounts	High
S4	SSH weak, verified.	High	
	sshd -T: passwordauthentication yes (set in /etc/ssh/sshd_config.d/50-cloud-init.conf), permitrootlogin without-password (root can log in with a key), maxauthtries 6. fail2ban inactive. Port 22 on 0.0.0.0. Login+sudo password for tm is tm (2 chars). 4 keys in ~/.ssh/authorized_keys.		
S5	NFS /mnt/storage exported RW, no auth, to 192.168.0.0/24	/etc/exports: rw, sync, all_squash, anonuid=1000, anongid=1000, insecure, fsid=0	Medium
S6	Samba shares /mnt (whole tree) RW; [homes] browsable; guest mapping on	testparm: [mnt_tmshare] path=/mnt read only=No; usershare allow guests=Yes; map to guest=Bad User	Medium
S7	arr apps have no auth from LAN	AuthenticationRequired=DisabledForLocalAddresses on Radarr/Sonarr/Prowlarr	Medium
S8	~25 service ports on 0.0.0.0	incl. Immich 2283, Immich Power-Tools 8001, seerr 5055, qbt 8080, radarr 7878, sonarr 8989, prowlarr 9696, searxng 8088, litellm 4000, gpt-researcher 8000, home-dash2 3000, baileys 3002, prometheus 9090, alertmanager 9093	Medium
S9	Cloudflare tunnel token in .env and process args	docker inspect cloudflared exposes the full token to any local user	Medium
S10	glancesweb.service ran Glances web UI as root on 0.0.0.0	now failed (ModuleNotFoundError: No module named 'defusedxml' — broken pip install), but still enabled — retries on every boot	Low (currently)
S11	Pending reboot for linux-image-6.8.0-139 (running -138)	reboot-required.pkgs. (Auto-updates are working — not a finding.)	Low–Med
S12	tm / tm everywhere	Same 2-char string is the SSH login password, the sudo password, and TERRAMASTER_PASSWORD in .env (NAS web UI).	High

#	Finding	Detail / evidence	Severity
S13	Single user, no MFA anywhere ; tm is in sudo + docker + lxd	id: sudo,docker,lxd,render — docker group = root-equivalent	Info
S14	Immich has no offsite/backup for originals	268 GB on one mergerfs disk, no parity	Medium (data-loss, not intrusion)
S15	No UPS; 35 unclean NVMe shutdowns	lsusb / no nut; nvme smart-log unsafe-shutdown counters 21 + 14	Medium-High (data integrity)
S16	No SMART self-test schedule; alerts go to local root mail	/etc/smartd.conf has no -s schedule; -m root only	Medium (on a no-parity pool)

Good, for balance: no privileged containers; no `docker.sock` mounts; `iptables FORWARD` policy is `DROP` (Docker-managed) and the Tailscale ACL chain correctly drops tailnet source addresses arriving on non-tailscale0 interfaces; Tailscale has **no funnel/serve** (nothing exposed to the internet via Tailscale); only `theownitguy-website` is public (via Cloudflare, isolated on its own bridge network); qBittorrent is private-tracker-safe (DHT/PeX/LSD off); `unattended-upgrades` is on; `fstrim.timer` and `e2scrub_all.timer` run weekly; the trading-platform stack is genuinely well-hardened; Immich DB is backed up nightly with 14-day retention; **all 6 pool drives are SMART-clean with zero reallocated/pending sectors.**

13. Legacy / orphaned inventory

Item	Evidence	Action
Exited containers: <code>signen-web</code> , <code>signen-scheduler</code> , <code>signen-baileys</code> (project v2)	Exited (137) 12 days ago	<code>docker compose -p v2 down</code>
Exited containers: <code>life360-webapp</code> , <code>life360-poller</code> (project <code>life360-nas-poller</code>)	Exited (0) 11 days ago — replaced by <code>home-dash2</code> 's poller	<code>docker compose -p life360-nas-poller down</code>
38.3 GB reclaimable images (97 % of image store)	<code>docker system df</code>	<code>docker image prune -a</code>
14.9 GB reclaimable build cache	<code>docker buildx du</code>	<code>docker builder prune</code>
Unused images: <code>v2-signen-*</code> , <code>home-dash2-staging-*</code> , <code>homedash2-signen-scheduler</code> , <code>signen-map-test-*</code> , <code>home-dash2-life360-*</code> , <code>life360-nas-poller-*</code>	<code>docker images vs docker ps -a</code>	<code>prune</code>
Orphan networks: <code>homedash2_default</code> , <code>v2_default</code> , <code>life360-nas-poller_default</code>	<code>docker network ls</code>	<code>docker network prune</code>
<code>glancesweb.service</code> — enabled, failed, crash-looped	<code>systemctl status glancesweb</code>	<code>sudo systemctl disable --now glancesweb && sudo rm /etc/systemd/system/glancesweb.service</code> delete (nightly gz backups exist)
<code>/mnt/immich_db_backup.sql</code> — 4.4 GB, 2026-07-04	<code>ls -la /mnt</code>	
<code>~/signen-dashboard/_legacy_v1_unused_data/history.db</code> — 195 MB		keep only if it has history you still want
<code>~/signen-dashboard/{v2,migration-test,data-staging,backups}</code>		archive to a tarball, remove

Item	Evidence	Action
~/trading-platform-*.bundle × 9 (~1.5 MB total)	ls ~/.bundle	delete — the GitHub remote is the backup now
~/*.py / ~/*.sh loose scripts (check*.py, console*.js, topup.py, modbus.py, deploy_unpoller*.sh, extract_devices*.sh, get_devices.sh, unifi_api_client.py, ...)	ls ~	move to ~/scratch/ or a misc-scripts repo
~/config/{plex,ntopng} empty; ~/config/overseerr vs container name seerr	du -sh ~/config/*	tidy naming
.DS_Store and .* AppleDouble files throughout /mnt/storage, ~/sigen-dashboard	find	find ... -name '.*' -o -name .DS_Store -delete; set fruit:veto_appledouble
~/sigen-dashboard/{sigen-cron.log, sigen -logger.log}		delete
/mnt/{das1,das2,scratch,usb_data,remote_usb,vmmain1p2,ats,atuk,temp}	rm -rf /mnt/{das1,das2,scratch,usb_data,remote_usb,vmmain1p2,ats,atuk,temp}	consolidate to one /mnt/scratch
autobrr container — 0 config	autobrr.db all-zero	configure or docker compose remove
Plex “F1” library extra path	section_locations	remove that path in Plex
/mnt/NVMe2		
qBittorrent.conf	contradicts	reconcile to one path
Preferences\Downloads\SavePath=/mnt/Downloads\DefaultSavePath	sed -i s/Downloads/DefaultSavePath/	

14. GitHub review — what is (and isn’t) replicated

No gh CLI, no ~/.gitconfig. GitHub account: **alc168** (github.com/alc168). Auth is via SSH keys, including a dedicated host alias github-trading-platform (a per-repo deploy key).

14.1 Repos on the box

Local path	Remote	Branch	Working tree	On GitHub?
~/sigen-dashboard/home to github	git@github.com:alc168/sigen-dashboard.git	cbfe0ef	clean	Yes — origin/main present; HEAD not ahead of it
~/trading-platform	git@github-trading-platform:alc168/trading-platform.git	main	clean	Yes — has CI (.github/), PR #1 merged
~/hybrid-research	github.com/alc168/hybrid-research.git	main	clean	Yes
~/life360-nas-poller	github.com/alc168/life360-nas-poller	main	32 modified files, uncommitted	Yes, but local has drifted; project is retired
~/territory	/home/tm/git/territory.git (local bare repo only)	main	clean	No — not on GitHub. Has RECOMMENDATIONS.md from a 2026-09-06 review
~/mergerfs-tools	github.com/trapexit/mergerfs-tools.git	master	clean	third-party clone

14.2 What is NOT in version control at all

Everything that actually defines this server:

- ~/docker-compose.yml (the whole media + Immich + Plex stack)
- ~/docker-compose-monitoring.yml

- ~/config/* (Radarr/Sonarr/Prowlarr/qBittorrent/Plex/Grafana/Prometheus/...)
- ~/research-agent/, ~/theownitguy-website/, ~/timeline-nas-webapp/, ~/vce-psychology/
- ~/sigen-dashboard/ top level (only the home.dash2/ subdir is a repo)
- /etc/nginx/sites-*, /etc/exports, /etc/samba/smb.conf, /etc/fstab, the systemd units, crontab, ~/scripts/

If the OS SSD dies you can rebuild the *data* (mostly) but you would be reconstructing the *configuration* from memory. See Recommendation 1.

14.3 Documentation that exists

~/sigen-dashboard/ is heavily documented (README 30 KB, AGENTS.md, ARCHITECTURE.md, a 56 KB Architecture-Reference-Proposal, tutorials, a Home-Network-Security-Report .md+.pdf, plus docs/ duplicates and ._* AppleDouble copies). home.dash2/README.md is a genuine self-contained tutorial. trading-platform has docs/ and CI. Nothing else has a README.

15. Recommendations — one by one

Ordered: **A** = security (do first, ~1 evening), **B** = robustness / backup, **C** = cleanup, **D** = filesystem re-sort. Each is independent unless noted.

A. Security

A1 — Change the Grafana admin password now. ~/.env: set GF_SECURITY_ADMIN_PASSWORD to a real secret, then cd ~ && docker compose -f docker-compose-monitoring.yml up -d grafana. Log in, delete the old default. (2 min, removes the single worst finding.)

A2 — Put a host firewall in front of everything.

```
sudo apt install ufw
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow from 192.168.0.0/24 to any port 22,80,443,445,2049,32400,2283,5055,3000,3001 proto tcp
sudo ufw allow in on tailscale0
sudo ufw enable
```

Then decide, service by service, which of the ~25 open ports actually need to be on the LAN vs. only reachable over Tailscale. Anything that is “just for me” (Prometheus 9090, Alertmanager 9093, litellm 4000, gpt-researcher 8000, power-tools 8001, baileys 3002, qbt 8080, radarr/sonarr/prowlarr) should be Tailscale-only.

A3 — Harden SSH and change the tm password. First: sudo passwd tm -> a real passphrase (it is currently tm, and that is also your sudo password). Then, because /etc/ssh/sshd_config.d/50-cloud-init.conf sets PasswordAuthentication yes, override it in /etc/ssh/sshd_config.d/99-hardening.conf (later file wins):

```
PasswordAuthentication no
KbdInteractiveAuthentication no
PermitRootLogin no
AllowUsers tm
```

sudo systemctl restart ssh (verify with sudo sshd -T | grep -E 'passwordauthentication|permitrootlogin'). You already authenticate with a key. Then either install fail2ban (sudo apt install fail2ban, default jail is fine) or, better, stop exposing 22 to the LAN at all and reach the box over Tailscale SSH.

A4 — Rotate the secrets in §11, starting with the ones that are reused personal passwords (SIGEN_CLOUD_PASSWORD, MEROSS_PASSWORD) and TERRAMASTER_PASSWORD=tm. Give the NAS admin account a real password. Move secrets out of .env files into either Docker secrets (as the trading-platform stack already does) or at minimum chmod 600 + .gitignore verified for every .env.

A5 — Turn on auth for the arr apps. Set AuthenticationRequired=Enabled (not DisabledForLocalAddresses) in each config.xml, or front them with a single authenticating reverse proxy. Same for qBittorrent WebUI (it already has a password; make sure WebUI\Address is not * once behind the proxy).

A6 — Lock down NFS and Samba. NFS: export read-only, and to specific host IPs, not the whole /24 — e.g. /mnt/storage 192.168.0.50(ro,sync,no_subtree_check,root_squash). Samba: scope [mnt_tmshare] to a specific subtree (e.g. /mnt/storage) instead of all of /mnt, set usershare allow guests = No, and confirm map to guest is intentional.

A7 — Reboot the box into linux-image-6.8.0-139 (pick a low-use window; `sudo reboot`). `unattended-upgrades` is already enabled, so this is the only outstanding update action. Consider adding `Unattended-Upgrade::Automatic-Reboot "true"`; with a 04:00 window in `/etc/apt/apt.conf.d/50unattended-upgrades` so kernel patches don't sit unbooted for weeks.

A8 — Disable the dead Glances unit so it does not resurrect on reboot: `sudo systemctl disable --now glancesweb.service && sudo rm /etc/systemd/system/glancesweb.service`.

A9 — Cloudflare tunnel: confirm the Zero-Trust dashboard only routes the one hostname to theownitguy and that no path maps back to other NAS ports. Regenerate the tunnel token (it has been on disk in cleartext and in `docker inspect`).

B. Robustness & backup

B1 — Put the server config in git. Create `alc168/nas-config` (private). Track: `~/docker-compose*.yaml`, every app dir's `docker-compose.yaml` + `*.example` env (never the real `.env`), `~/scripts/`, and copies of `/etc/nginx/sites-available/*`, `/etc/exports`, `/etc/samba/smb.conf`, `/etc/fstab`, `crontab -l`, the `systemd` unit files. A `Makefile` or `bootstrap.sh` that recreates the layout. This is the single highest-value robustness change.

B2 — Fix the home.dash2 detached HEAD:

```
cd ~/sigen-dashboard/home.dash2
git checkout main && git pull --ff-only
```

and confirm `cbfe0ef` is already on `origin/main` (it appears to be). Add a pre-deploy check that refuses to build from a detached HEAD.

B3 — Commit or discard the 32 dirty files in ~/life360-nas-poller, then —since the project is retired—archive the repo on GitHub and `docker compose -p life360-nas-poller down --rmi local`.

B4 — Push territory to GitHub (it only exists as a local bare repo at `~/git/territory.git` — one disk failure from gone).

B5 — Protect the Immich originals. 268 GB is small enough for an inexpensive offsite: `rclone` to a cloud bucket (Backblaze B2 / Storj) on a weekly cron, or an external USB drive rotated monthly. The nightly DB dump is worthless without the photos.

B6 — Add a restic/borg job for the irreplaceable small stuff: `~/sigen-dashboard/data/history_v2.db`, `~/config/*` (minus Plex's 31 GB Library/), the trading Postgres, `territory.db`, `vce-data`. Target `/mnt/storage/backups/` and offsite.

B7 — Right-size logging. Set `LogLevel` back to `info` on Radarr, Sonarr, Prowlarr (reclaims most of the 2.3 GB / 422 MB / 42 MB). Add logging: `{driver: json-file, options: {max-size: 10m, max-file: 3}}` to every service in `~/docker-compose.yaml` (the trading and home-dash2 stacks already do this).

B8 — Add swap headroom or RAM. 7.5 GiB with this many resident services means constant swapping. Either bump the box to 16 GiB (an N95 typically takes a single SO-DIMM up to 16 GiB) or shut down the stacks you rarely use (see C1).

B9 — Consider parity for the media pool. `mergerfs` + **SnapRAID** gives you one- or two-disk parity for the 47 TB of media without converting to a real RAID. Even one parity disk turns “lost a disk = lost those movies” into “lost a disk = rebuild”. A weekly `snapraid sync` + monthly `snapraid scrub` timer also gives you bit-rot detection the pool currently has none of.

B10 — Buy a small UPS and wire up NUT. The NVMe SMART logs show **35 unclean shutdowns**. A ~600–900 VA UPS + `sudo apt install nut`, configured for a clean shutdown at ~50 % battery, protects the two Postgres instances, the dozen SQLite DBs, and the ext4 filesystems from the power cuts that are already happening. This is the highest-value robustness spend on the box.

B11 — Schedule SMART self-tests and real alerting. Append to `/etc/smartd.conf` (then `sudo systemctl restart smartmontools`):

```
/dev/sda -a -o on -S on -s (S/../.././02|L/../../6/03) -W 4,45,55 -m <you@email> -M exec /usr/share/smartmontools/smartmontools.conf
# ...one line per /dev/sd[a-f] and /dev/nvme[01]...
```

(short test nightly 02:00, long test Saturday 03:00, temperature warnings). Point `-m` at an address that reaches you, or a script that pings the Alertmanager you already run. Add a Prometheus `smartctl_exporter` / `node_exporter` `--collector.smartmon` panel so drive health shows up in Grafana next to everything else.

B12 — Pre-empt disk2 (sdb, 45,600 h / 5.2 yr). It is SMART-clean but statistically the first to go, and the pool has no parity. Either buy a cold spare now, or plan a `mergerfs` drain of that disk (`mergerfs.balance` / manual `rsync` off, then swap) at the next maintenance window. While you are in there, note `disk1` (sda) is an **SMR** drive — fine for cold media, bad as a torrent-write target; steer new `category.create` writes away from it or replace it.

C. Cleanup (safe, reversible)

C1 — Decide which stacks are keepers. Running today but arguably idle: `research-agent` (3 containers, 4.7 GB `gpt-researcher` image), `territory`, `timeline-nas-webapp`, `bitmappery` (3.15 GB image), `theownitguy-website`. Each one you stop frees RAM and a port. `docker compose -p <name> stop` is instant and reversible.

C2 — Prune Docker. After C1:

```
docker compose -p v2 down
docker compose -p life360-nas-poller down
docker container prune
docker image prune -a          # ~38 GB
docker builder prune          # ~15 GB
docker network prune
```

Expect ~50 GB back on /.

C3 — Delete the stray dumps & bundles. `rm /mnt/immich_db_backup.sql` (4.4 GB), `rm ~/trading-platform-*.bundle`, `rm ~/sigen-dashboard/{sigen-cron.log,sigen-logger.log}`.

C4 — Collapse ~/sigen-dashboard. Keep `home.dash2/`, `data/`, `.env`, `life360.env`, and *one* copy of the docs (in the `home.dash2` repo). Tar the rest (`_legacy_v1_unused`, `v2`, `migration-test`, `data-staging`, `backups`, `loose root *.md/*.pdf`) to `~/archive/sigen-dashboard-legacy-2026-09.tar.zst` and delete the originals. Extract `_legacy_v1_unused/history.db` first if you want the old history.

C5 — Sweep AppleDouble litter.

```
find /mnt/storage ~/sigen-dashboard -name '.*' -o -name '.DS_Store' -delete
```

and set `fruit:veto_appledouble = yes` in `smb.conf` to stop it recurring.

C6 — Tidy loose ~ scripts into `~/scripts/` (or a `misc-scripts` repo) and out of `$HOME`: `check*.py`, `console*.js`, `costs.py`, `investigate.py`, `modbus.py`, `row.py`, `sim_test.py`, `topup.py`, `deploy_unpoller*.sh`, `extract_devices*.sh`, `get_devices.sh`, `enhanced_unifi_client.py`, `simple_unifi_client.py`, `unifi_api_client.py`, `add_monitoring.py`, `create_compose.sh`, `device_summary.sh`, `fix_unpoller_endpoint.sh`, `unpoller_dreamwall.sh`.

C7 — Remove `docker-compose*.yaml.bak*` / `.backup` from ~ once B1 is done (git is the history now).

C8 — Either configure autobrr or remove it. If you want instant TL grabs: add the TorrentLeech IRC network + announce channel, a filter, and a qBittorrent action. Otherwise delete the service from `~/docker-compose.yml`.

C9 — Fix the Plex “F1” library — remove the `/mnt/NVMe2` path (Settings -> Libraries -> F1 -> Edit -> remove folder).

D. Filesystem re-sort

D1 — Move Immich onto the NVMe it was bought for. `/mnt/NVMe2` (932 GB, 2 % used) is literally labelled “for Photos” in `/etc/fstab`. Plan: 1. `docker compose stop immich-server immich-machine-learning immich-microservices database redis` 2. `rsync -aH /jez-cache/immich/ /mnt/NVMe2/immich/` (DB data + model cache) 3. Move the 268 GB of originals: `rsync -aH --info=progress2 /mnt/storage/jez-photos/ /mnt/NVMe2/immich/upload/` (268 GB + thumbs/encoded fits comfortably in 932 GB) 4. Repoint the bind mounts in `~/docker-compose.yml` (`/mnt/NVMe2/immich/upload:/usr/src/app/upload`, `/mnt/NVMe2/immich/postgres:/var/lib/postgresql`, `/mnt/NVMe2/immich/model-cache:/cache`) and set `DB_STORAGE_TYPE='SSD'`. 5. `docker compose up -d`, verify, then

remove the old dirs. Result: photo scrubbing/thumbnails/ML get NVMe random-IO, the OS SSD gets ~3 GB back, and the mergerfs pool gets ~300 GB back.

D2 — Give Plex its own config root. Change the compose mount from `./config:/config` to `./config/plex:/config` (or a dedicated `~/plex-config`), migrate `~/config/Library/` -> the new location, so a backup or chown of `~/config` no longer drags 31 GB of Plex thumbnails and Plex is not sharing a directory with six other apps.

D3 — Reconcile the torrents area (the 7.6 TB). In Radarr and Sonarr enable **Settings -> Media Management** -> “Use Hard Links instead of Copy”. Then audit `/mnt/storage/torrents`:

```
# entries with only one hardlink = not linked into the library = candidates
find /mnt/storage/torrents -type f -links 1 -size +100M -printf '%k KB\t%p\n' | sort -rn | head -50
```

Cross-check against qBittorrent’s active-seeding list. Anything not seeding for ratio and already in the library can go. Set a qBittorrent **share-limit** (ratio or seed-time) with “remove torrent and files” so this does not rebuild.

D4 — One scratch dir. Replace `/mnt/{temp,scratch,usb_data,remote_usb,nvme1n1p2-scratch}` and `/mnt/das1`, `/mnt/das2` with a single `/mnt/scratch` (and delete the commented DAS lines from `/etc/fstab` if the D4-320 enclosure is gone).

D5 — Normalise the download taxonomy. qBittorrent has new `movies`, new `tv`, `tv`, `radarr`, `tv-sonarr` categories pointing at four different paths (`/mnt/storage/data/movies`, `/mnt/storage/data/tv`, `/mnt/storage/torrents`). Pick one convention: `radarr -> /mnt/storage/torrents/movies`, `tv-sonarr -> /mnt/storage/torrents/tv`, delete the rest, and update `Session\DefaultSavePath` and the stale `Preferences\Downloads\SavePath` to match.

D6 — Move ~/config app data that is really *state* off the OS SSD only if / gets tight — at 12 % it is fine for now. The bigger win is D1.

Appendix A — full container list (running)

```
IMMICH:      immich_server(2283) immich_machine_learning immich_postgres
             immich_redis immich_power_tools(8001)
MEDIA:       plex2(host) qbt-tl(host:8080) sonarr(host:8989) radarr(host:7878)
             prowlarr(host:9696) autobrr(host:7474) seerr(5055) kometa
MONITORING:  prometheus(9090) grafana(3001) alertmanager(9093)
             unpoller(9130) node_exporter
HOME-DASH2:  home-dash2-sigen-web-1(3000) home-dash2-sigen-scheduler-1
             home-dash2-sigen-baileys-1(3002) home-dash2-life360-api-1
             home-dash2-life360-poller-1
TRADING:     trading-platform-{postgres,api(8092),ui(8093),collector(9108),
             sentiment(9109),macro(9110),equity,maintenance}-1
WEB APPS:    territory-{frontend(8802),backend(8801)}-1 vce-psychology(8137)
             theownitguy-website cloudflared timeline-webapp(4002)
             bitmappery(5173)
RESEARCH:    gpt-researcher(8000) searxng(8088) litellm(4000)
```

Appendix B — evidence files

Raw command output captured during the audit is in the working directory alongside this report (`01-disk-mounts ... 15-photos`). Key sources: `docker ps -a`, `docker compose ls -a`, `docker inspect`, `docker system df`, `/etc/fstab`, `/etc/exports`, `testparm -s`, `ss -tulpn`, `crontab -l`, `~/docker-compose.yml`, `~/docker-compose-monitoring.yml`, `~/sigen-dashboard/home.dash2/docker-compose.yml`, `prowlarr.db` / `radarr.db` / `sonarr.db` / `autobrr.db` queries, `com.plexapp.plugins.library.db` queries, Immich Postgres queries, `git remote -v` across all repos.